Hire-Smart ATS
AI-Based Resume Screening & Feedback Platform

Project Report

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Submitted by:
Aditya Gupta
B.Tech CSE (AI&DS)
IVth Year, VIIIth Sem.
SVVV, Indore

Submitted to:
Mr. Rushikesh Chavhan Sir
Data Analyst, IBM

Aditya Gupta

Mr. Rushikesh Chavhan

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Abstract

HireSmart ATS is a full-stack AI-powered resume screening and feedback platform designed to assist users in evaluating and improving their resumes using both rule-based NLP techniques and machine learning. The system provides two modes of analysis: Job Description (JD) based resume evaluation and a General ATS Score prediction using a trained regression model. Users can upload resumes and job descriptions in PDF format or paste job descriptions as text, after which the system automatically extracts and processes the content. For JD-based analysis, the application computes an ATS Match Score, Resume Rating, identifies matched and missing skills, and generates actionable suggestions to improve resume relevance. In parallel, a machine learning regression model trained on an ATS score dataset predicts a continuous General ATS Score, representing the overall quality of the resume on a percentage scale.

The machine learning pipeline utilizes TF-IDF for text feature extraction and a regression model to produce realistic, fine-grained scoring instead of fixed or rule-based outputs. The trained model is deployed using FastAPI as a RESTful service and is seamlessly integrated with a modern React and TypeScript frontend. The platform features a clean, responsive dashboard for result visualization, enabling real-time feedback with minimal latency. This project demonstrates the practical application of Natural Language Processing, Machine Learning model training and deployment, and full-stack web development in a real-world recruitment and Applicant Tracking System (ATS) context.

1. Introduction

In modern recruitment processes, Applicant Tracking Systems (ATS) play a crucial role in filtering and ranking resumes before human review. Many candidates are rejected due to poor ATS compatibility rather than lack of skills. This project aims to build an intelligent web-based system that helps users evaluate and improve their resumes by simulating an ATS-like evaluation process.

HireSmart ATS provides two modes of analysis:

1. **Job Description (JD) Based Analysis -** Evaluates how well a resume matches a specific job description.
2. **General ATS Score (ML-Based) -** Predicts the overall quality of a resume using a trained machine learning regression model.

2. Objectives

- To build a web application for resume analysis using AI and NLP techniques.
- To implement PDF text extraction for resumes and job descriptions.
- To compute ATS Match Score, Resume Rating, and skill gap analysis.
- To train and deploy a machine learning regression model for predicting a continuous General ATS Score.
- To integrate the ML backend with a modern frontend dashboard.
- To provide actionable suggestions for improving resumes.

3. System Overview

The system consists of three main components:

1. **Frontend (React + TypeScript)**
    - User interface for uploading PDFs and viewing results.
2. **Backend ML API (FastAPI)**
    - Hosts the trained ML model and exposes a `/predict` endpoint.
3. **Machine Learning Training Pipeline**
    - Trains a regression model on an ATS score dataset and exports the model for inference.

Workflow:

1. User uploads Resume PDF and optionally Job Description PDF or text.
2. System extracts text from PDFs.
3. JD-based NLP analysis computes match score, rating, skills, and suggestions.
4. Resume text is sent to ML API for General ATS Score prediction.
5. Results are displayed in a dashboard UI.

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4. Technology Stack

Frontend

- React + TypeScript
- Vite
- Tailwind CSS
- Framer Motion
- shadcn/ui components

Backend

- Python
- FastAPI
- Uvicorn

Machine Learning / NLP

- Scikit-learn
- TF-IDF Vectorizer
- Regression Model (Linear/Ridge Regression)
- Joblib
- NumPy

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5. Requirements Analysis

The primary requirements of the system were identified as follows:

Functional Requirements:

- Upload Resume in PDF format.
- Upload Job Description either as text or PDF.
- Extract text from uploaded PDFs.
- Perform Job Description based resume analysis.
- Compute ATS Match Score and Resume Rating.
- Identify matched and missing skills.
- Provide suggestions to improve the resume.
- Predict a General ATS Score using a Machine Learning model.
- Display all results in a user-friendly dashboard.

Non-Functional Requirements:

- The system should be responsive and easy to use.
- The ML prediction should be fast and reliable.
- The system should handle PDF files efficiently.
- The architecture should be modular and scalable.

These requirements guided the design and implementation of the system.

Software Requirements:

- Python 3.x
- Node.js and npm
- Web Browser (Chrome/Edge/Firefox)

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6. Project Structure

```
HireSmart-ATS/
├── frontend/
│   └── src/
│       ├── App.tsx
│       ├── components/
│       │   ├── UploadForm.tsx
│       │   └── ResultsDashboard.tsx
│       ├── lib/
│       │   ├── nlp.ts
│       │   └── pdf-extract.ts
├── ml_api/
│   ├── ml_api.py
│   ├── general_ats_regressor.pkl
│   ├── tfidf_vectorizer.pkl
│   └── requirements.txt
├── training/
│   ├── train_general_ats_regression.py
│   ├── train.csv
│   └── validation.csv
└── README.md
```

- **Frontend Layer (React + TypeScript):**
Handles user interaction, file uploads, and result visualization.

- **Backend Layer (FastAPI):**
Hosts the trained ML model and exposes REST endpoints for prediction.

- **Machine Learning Layer:**
Contains the regression model trained using TF-IDF features and an ATS score dataset.

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7. Machine Learning Methodology

7.1 Dataset

The model is trained on a Resume ATS Score Dataset containing text samples and corresponding continuous ATS scores.

7.2 Preprocessing

- Text cleaning (lowercasing, whitespace normalization)
- Feature extraction using TF-IDF Vectorizer
- Use of n-grams to capture keyword patterns

7.3 Model

- A regression model (Linear/Ridge Regression) is used.
- Input: TF-IDF feature vector of resume text
- Output: Continuous score between 0 and 100

7.4 Training and Evaluation

- Model is trained on training data and evaluated on validation data.
- Performance metrics such as Mean Absolute Error (MAE) and R^2 score are used.
- The trained model and vectorizer are saved using Joblib.

7.5 Inference

- The FastAPI backend loads the saved model and vectorizer.
- Endpoint: `POST /predict`
- Returns:

```
{ "score": 59.29, "label": "Average" }
```

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8. Backend Implementation (FastAPI)

- The backend exposes a REST API endpoint `/predict`.
- It accepts resume text as input and returns a predicted ATS score.
- CORS is enabled to allow frontend communication.
- The backend serves as a model inference service.

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9. Frontend Implementation

9.1 UploadForm Component

- Allows uploading Resume PDF.
- Allows pasting JD text or attaching JD PDF.
- Extracts text from PDFs.
- Calls:
  - NLP functions for JD-based analysis.
  - ML API for General ATS Score.

9.2 ResultsDashboard Component

- Displays:
  - Resume Rating (out of 10)
  - ATS Match Score (%)
  - General ATS Score (%)
  - Matched Skills
  - Missing Skills
  - Suggestions
- Uses cards, progress bars, and a responsive layout.

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10. Working of the System

1. User uploads Resume PDF.
2. User optionally uploads JD PDF or pastes JD text.
3. System extracts text from PDFs.
4. JD-based NLP analysis computes:
   - ATS Match Score
   - Resume Rating
   - Matched and Missing Skills

    - Suggestions
5. Resume text is sent to ML API.
6. ML model predicts a continuous General ATS Score.
7. Results are displayed on the dashboard.

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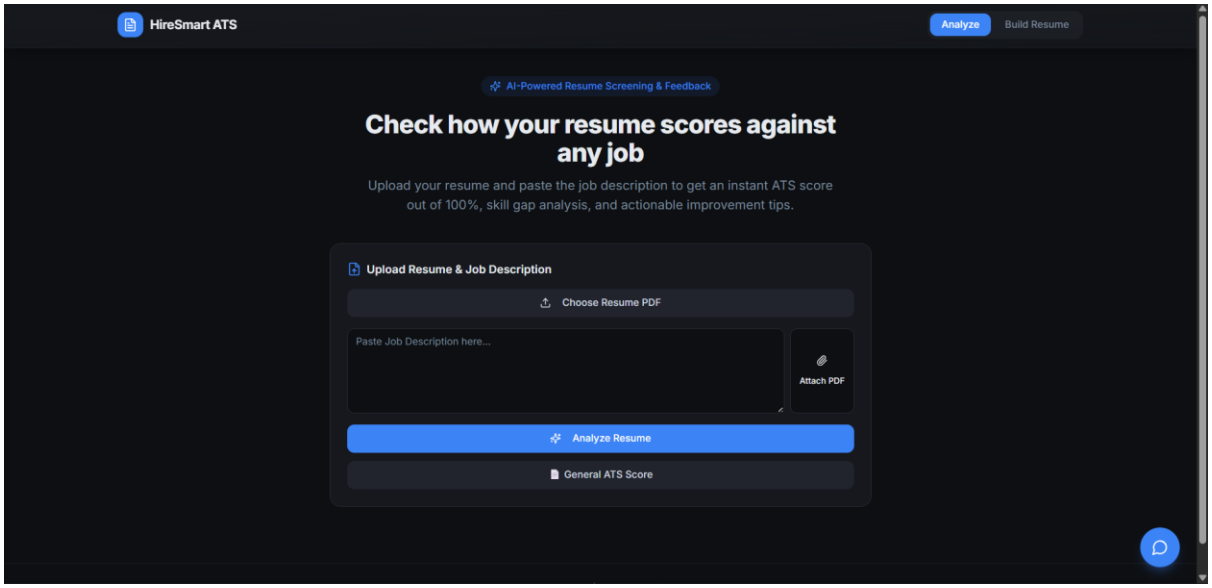
11. Results and Output

- The system produces:
    - ATS Match Score (percentage)
    - Resume Rating (out of 10)
    - General ATS Score (continuous ML-based percentage)
    - Skill gap analysis and suggestions
- Different resumes produce different scores, demonstrating the effectiveness of the regression model.



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12. Advantages

- Provides real-time resume evaluation.
- Uses machine learning for realistic continuous scoring.
- Helps users improve resumes for ATS systems.
- Demonstrates full-stack AI system integration.

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13. Limitations

- Model accuracy depends on the quality and size of training data.
- NLP-based skill extraction is rule-based and can be improved with advanced models.
- The system currently focuses on text-based resumes only.

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14. Future Scope

- Integration of advanced NLP models (e.g., BERT).
- Support for more file formats (DOCX).
- Improved skill extraction using named entity recognition.
- User authentication and resume history tracking.
- Cloud deployment for public access.

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15. Conclusion

HireSmart ATS successfully demonstrates the application of NLP and machine learning in building an intelligent resume screening system. By combining JD-based analysis with an ML-based regression model for general resume quality scoring, the system provides both targeted and holistic feedback. This project showcases an end-to-end AI solution including data processing, model training, API deployment, and frontend integration.

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16. References

- Scikit-learn Documentation
- FastAPI Documentation
- React Documentation
- ATS Score Dataset (Hugging Face / Kaggle)
- Research articles on Resume Screening and NLP